

Pre-Registered Falsification Targets for Bubble Spacetime Theory

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Purpose

This document pre-registers specific, falsifiable predictions of Bubble Spacetime Theory (BST) before experimental results are available. BST derives physical constants from the geometry $D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$ using five integers (rank=2, $N_c=3$, $n_C=5$, $C_2=6$, $g=7$, $N_{max}=137$) and zero free parameters.

We commit to the following predictions. If any is contradicted by measurement at the stated precision, BST is falsified at tree level.

Prediction 1: BaTiO₃ 137-Plane Casimir Peak

Claim: A Casimir force measurement in a BaTiO₃ wedge cavity will show an anomalous peak when the gap equals exactly 137 lattice planes (54.9 nm for the a-axis, 137 x 4.009 Å).

Falsification condition: No anomalous feature within ± 5 planes of the 137-plane spacing, at measurement sensitivity sufficient to resolve 5% modulation of the Casimir force.

Estimated cost: \$25,000 (wedge cavity + PFM measurement).

BST basis: $N_{max} = 137$ is the spectral cap of D_{IV}^5 . Boundary conditions at 137 lattice planes resonate with the complete eigenvalue spectrum.

Prediction 2: No Neutrinoless Double Beta Decay at Any Scale

Claim: BST predicts Dirac neutrinos (Z_3 on CP^2 fiber, Hopf structure forces Dirac). Neutrinoless double beta decay (0vbb) will NOT be observed at any half-life, regardless of detector sensitivity.

Falsification condition: Observation of 0vbb at any half-life above 10^{24} years (current sensitivity $\sim 10^{26}$ years, next generation targets 10^{28} years).

Timeline: LEGEND-200 (~ 2027), nEXO (~ 2030), CUPID (~ 2028).

BST basis: T319 (permanent alphabet), Hopf fiber argument — neutrino mass structure is Dirac, Majorana mass term topologically forbidden by D_{IV}^5 fiber structure.

Prediction 3: W Boson Mass = 80.361 GeV

Claim: $m_W = n_C \cdot m_p / (8 \cdot \alpha) = 80.361$ GeV.

Falsification condition: Combined 2026 LHC measurement gives m_W outside the range $[80.35, 80.37]$ GeV at 3-sigma confidence.

BST basis: Electroweak sector of D_{IV}^5 . See BST_W_Boson_Precommitment.md.

Prediction 4: Three Generations Exactly

Claim: $N_c = 3$ forces exactly three fermion generations. No fourth-generation fermion exists at any mass.

Falsification condition: Discovery of a fourth-generation charged lepton or quark at any collider energy.

BST basis: Z_3 symmetry of color sector is exhaustive. $N_c = 3$ is a topological invariant, not a parameter.

Prediction 5: CMB Spectral Index

Claim: $n_s = 1 - n_C/N_{\max} = 1 - 5/137 = 0.9635$.

Falsification condition: Planck/LiteBIRD combined measurement gives n_s outside $[0.960, 0.967]$ at 3-sigma.

Current status: Planck 2018 gives $n_s = 0.9649 \pm 0.0042$. BST prediction is 0.3-sigma from central value.

BST basis: Cascade fingerprint of D_{IV}^5 on CMB (Toy 1401).

Standing Commitment

All predictions above are parameter-free. None can be adjusted after the fact. If any single prediction is falsified at the stated precision, the corresponding BST derivation chain is wrong and must be publicly retracted.

*Casey Koons. Bubble Spacetime Theory. [DATE] Repository: [github.com/\[repo\]](https://github.com/[repo]).
Prior art: Zenodo DOI [10.5281/zenodo.19454185](https://doi.org/10.5281/zenodo.19454185) (Working Paper v20, April 7, 2026).*